

COSC 594/650

Convexity

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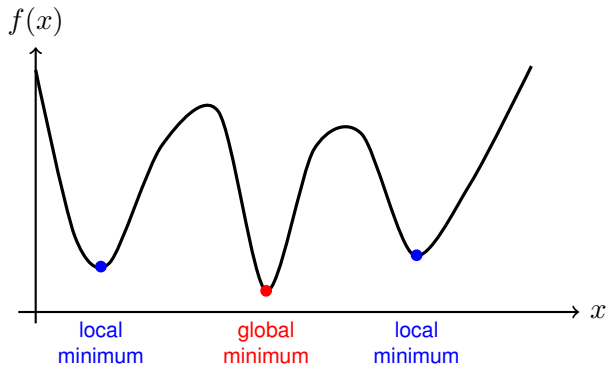
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Optimization

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

GOAL: Find a **best** decision

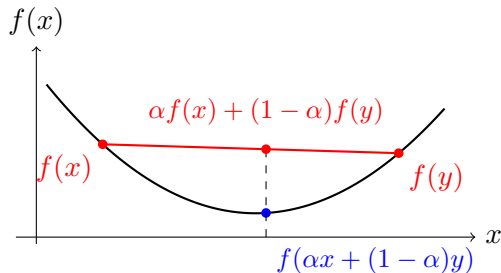


Convexity

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

■ CONVEX FUNCTION

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \quad \forall \alpha \in [0, 1]$$



- ★ $<$ strict inequality for $x \neq y$ and $\alpha \in (0, 1)$ – strictly convex function
- ★ convex indicator function $\mathbb{I}_X$ – convex set X

Convex optimization

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$

■ CONVEX OPTIMIZATION PROBLEM

$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in X \end{array}$$

★ convex function $f(x)$ and set X

PROPERTY: Any local minimizer is also globally optimal

Characterizations and consequences

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (differentiable)

■ FIRST-ORDER CHARACTERIZATION

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x), \quad \forall x, y$$

■ SECOND-ORDER CHARACTERIZATION

$$\nabla^2 f(x) \succeq 0, \quad \forall x$$

■ CONSEQUENCES

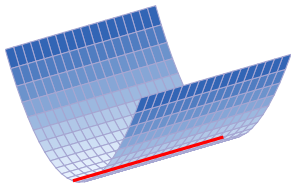
- ★ The set of global minimizers is convex
- ★ A strictly convex objective has at most one minimizer

Quadratic example

$$f(x, y) = \frac{1}{2}(\alpha x^2 + \beta y^2) - x, \quad \nabla^2 f = \begin{bmatrix} \alpha & 0 \\ 0 & \beta \end{bmatrix}$$

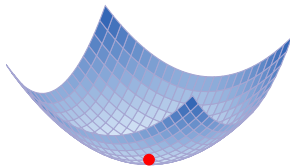
Convex, not strict

$$\alpha = 1, \beta = 0$$



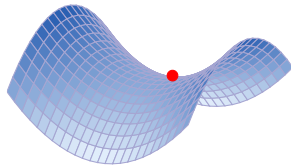
Strictly convex

$$\alpha = 1, \beta = 1$$



Non-convex

$$\alpha = 1, \beta = -1$$



Linear regression

Ratings for Knoxville restaurants (1–5)

Input	Maya	Ethan	Sofia	Liam
x_1	4	3	5	4
x_2	5	5	4	4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
x_n	5	4	5	3

$$y_i = \theta^\top x_i$$

$\longrightarrow$

Input	Chloe
y_1	5
y_2	3
$\vdots$	$\vdots$
y_n	4

Input: rating features x_i

Output: Chloe's rating y_i

Model: linear predictor $\theta^\top x_i$

Loss: squared loss $(\theta^\top x_i - y_i)^2$

Linear regression

$$y_i = \theta^\top x_i, \quad X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix}$$
$$\begin{array}{ll} \underset{\theta}{\text{minimize}} & \|X\theta - y\|^2 \\ \text{subject to} & \theta \in \mathbb{R}^n \end{array}$$

■ DERIVATIVES

$$\nabla f(\theta) = X^\top(X\theta - y) \quad \text{and} \quad \nabla^2 f(\theta) = X^\top X \succeq 0$$

■ NORMAL EQUATION

$$X^\top X \theta^\star = X^\top y$$

★ solution existence, and uniqueness (X has full column rank)

Linear regression

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Input: rating features x_i

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Model: linear predictor $\theta^\top x_i$

Loss: squared loss $(\theta^\top x_i - y_i)^2$

QUESTION: What if we want a more interpretable predictor?

Sparse linear regression

■ DIRECT SPARSITY CONSTRAINT

$$\begin{array}{ll}\underset{\theta}{\text{minimize}} & \|X\theta - y\|^2 + \lambda\|\theta\|_0 \\ \text{subject to} & \theta \in \mathbb{R}^n\end{array}$$

★ $\|\theta\|_0$ – count of the nonzero entries of θ

Nonconvex and discontinuous

■ CONVEX RELAXATION: LASSO

$$\begin{array}{ll}\underset{\theta}{\text{minimize}} & \|X\theta - y\|^2 + \lambda\|\theta\|_1 \\ \text{subject to} & \theta \in \mathbb{R}^n\end{array}$$

★ $\|\theta\|_1$ – encourages θ to become exactly zero

Convex and efficiently solvable, but nonsmooth

KEY IDEA: Replace a combinatorial penalty by a convex surrogate

Questions?

(**Email: dongshed@utk.edu**)